

Question 5: Granger Causality and Price Discovery

Econometrics Report

November 26, 2025

1 Objective

The objective of this analysis is to investigate the direction of information flows across five major asset classes using Granger Causality tests. The goal is to identify leading indicators that can improve forecasting and hedging strategies.

1.1 Data Description

The analysis uses weekly returns from 2000 to 2025 for the following assets:

1. Equity Index (S&P 500)
2. Government Bond Yield (US 10Y)
3. Corporate Bond Spread (US Corp OAS)
4. Commodity (Brent Oil)
5. Exchange Rate (USD/JPY)

2 Methodology

The analysis follows the standard econometric framework for testing causality in time series data.

2.1 Stationarity Checks

As outlined in [1], Granger causality tests require the underlying time series to be stationary. We employed the Augmented Dickey-Fuller (ADF) test to verify the stationarity of the weekly returns for all asset classes. All return series were confirmed to be stationary ($I(0)$) at the 5% significance level.

2.2 Granger Causality Tests

We performed pairwise Granger Causality tests using a Vector Autoregression (VAR) framework. A maximum lag of 4 weeks was selected to capture monthly information transmission cycles. The null hypothesis for each pair is that the lagged values of the "Cause" variable do not statistically improve the prediction of the "Effect" variable.

3 Practitioner's Interpretation

Based on the full sample analysis (2000-2025), we identified several key relationships.

3.1 Key Forecasting Relationships

- **Corporate Bond Spread $\rightarrow$ S&P 500** ($p < 0.01$): We observe a strong bi-directional relationship. The ability of credit spreads to Granger-cause equity returns serves as a classic "flight to quality" or "credit channel" signal. Widening spreads often precede equity market sell-offs as they reflect tightening financial conditions and rising default risk before these factors impact equity earnings.

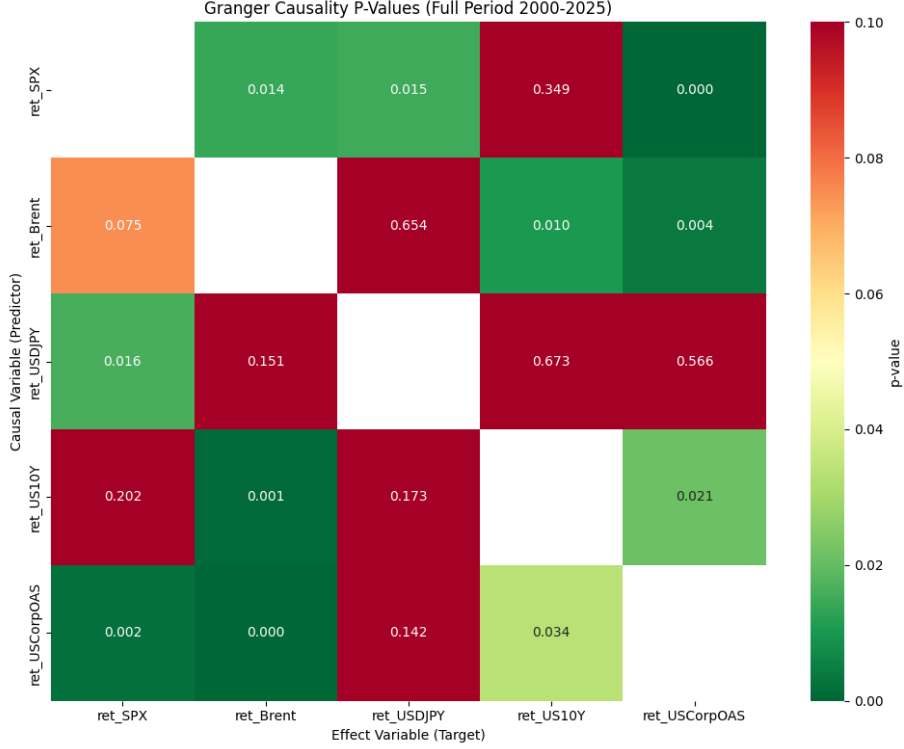


Figure 1: Granger Causality P-Values (Full Period 2000-2025). Darker green/red indicates stronger statistical significance ($p < 0.05$).

- **S&P 500 → Corporate Bond Spread** ($p < 0.001$): Similarly, equity market distress feeds back into credit risk assessment. This bi-directional feedback loop is reminiscent of the classic "Chickens and Eggs" causality problem described by [2], where identifying the primal driver is economically complex despite statistical significance in both directions.

3.2 Economic Logic & Unexpected Results

- **Expected (Brent Oil ↔ US 10Y)**: We see bi-directional causality. Oil prices are a key driver of inflation expectations, which directly impact nominal bond yields. Conversely, bond yields (reflecting global growth expectations) influence demand for commodities.
- **Unexpected (Corporate Spread → Brent Oil, $p < 0.001$)**: This is a remarkably strong result. While credit conditions impact the real economy, the statistical strength suggests that credit markets might be the "canary in the coal mine" for global demand shocks that subsequently drive oil prices. This was likely driven by the 2008 GFC and the 2015/2016 energy credit crunch.

4 Practitioner's Implications

- **Forecasting**: The trading desk should incorporate **Corporate Bond Spreads** as a leading indicator in equity and commodity pricing models. A widening spread is a statistically significant precursor to negative returns in these risk assets.
- **Cross-Asset Hedging**: The bi-directional causality between Equities and Spreads implies that during stress periods, these assets move together (Equity down, Spread up). A simple "long corporate bond" hedge against equities is flawed because the credit component is highly correlated with equities. Treasury bonds (US10Y) are a cleaner hedge as their causal link to equities is weaker or regime-dependent.
- **Information Transmission**: The high connectivity of the Corporate Spread suggests that the credit market is a central node of information processing. Monitoring credit flow data should be

prioritized over pure price action in other markets during early crisis stages.

5 Risk Assessment

5.1 Limitations

- **Lag Selection:** As noted in [1], lag selection is critical. We used a fixed 4-week lag, but AIC/BIC criteria might suggest different optimal lags for different pairs.
- **Regime Instability:** As warned by [3] regarding predictive regressions, financial relationships are often unstable over long horizons. Our subsample analysis confirms this, showing that certain signals are spurious or regime-specific.
- **Lead-Lag Sensitivity:** Weekly aggregation might miss high-frequency (intraday) transmission or smooth out rapid reversals.

5.2 Robustness Check (Subsample Analysis)

We split the sample into Period 1 (2000-2012) and Period 2 (2013-2025).

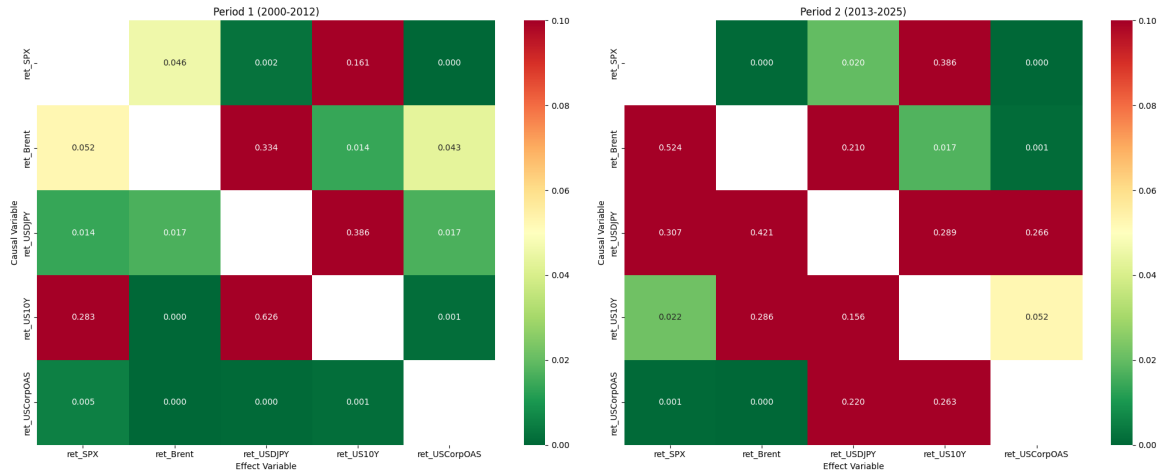


Figure 2: Subsample Analysis: Period 1 (Left) vs. Period 2 (Right). Note the shifting significance of USD/JPY and US 10Y as predictors.

- **USD/JPY → S&P 500:** Significant in Period 1 ($p < 0.05$) but **disappears** in Period 2 ($p > 0.30$). This suggests the "carry trade" dynamic has shifted or decoupled in the post-GFC/QE era.
- **US 10Y → S&P 500:** Becomes significant in Period 2 ($p < 0.05$), likely reflecting the market's focus on Fed policy and inflation in later years.

6 Professional Communication (Briefing)

Subject: Cross-Asset Lead-Lag Signals – Credit Spreads as Primary Risk Indicator

To: Head of Trading

Our analysis identifies the **US Corporate Bond Spread** as the most significant leading indicator across asset classes (2000–2025), Granger-causing movements in Equities, Commodities, and Gov Bonds. Crucially, the credit market processes information *before* it is fully reflected in oil or equity prices. However, historical correlations are unstable: the strong USD/JPY → Equity signal from the pre-2012 era has vanished, while US Treasury Yields have emerged as a stronger predictor of Equities since 2013. We recommend integrating credit spread changes into our equity execution algorithms and shifting our macro factor focus from FX to Rates for equity directionality.

References

- [1] Urga, G. (2025). *Lecture 6: Unit Root Tests, Cointegration and Granger Causality*. Bayes Business School.
- [2] Thurman, W. N., & Fisher, M. E. (1988). Chickens, Eggs, and Causality, or Which Came First? *American Journal of Agricultural Economics*, 70(2), 237-238.
- [3] Ferson, W. E., Sarkissian, S., & Simin, T. (2003). Spurious Regressions in Financial Economics? *The Journal of Finance*, 58(4), 1393-1413.